Sean Luka Girgis

PySpark / Hadoop Lead Data Engineer | SQL, Python, Shell & Data Quality

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LinkedIn | GitHub | Portfolio

Professional Summary

Senior Data Engineer with 20+ years of enterprise technology experience across Python, SQL, PySpark/Spark, Hadoop/Hive concepts, shell scripting, ETL/ELT pipelines, batch processing, data quality, data architecture, performance troubleshooting, and production reporting systems. Strong background building and validating large-scale data workflows, optimizing SQL and Spark-style processing, documenting transformation logic, and supporting regulated enterprise analytics environments. Best aligned to PySpark/Hadoop lead developer roles requiring SQL, Python, shell scripting, batch data processing, performance analysis, dimensional modeling, data quality, and enterprise delivery; Confluent Kafka, HBase, Impala, AutoSys, Elasticsearch, and Cloudera platform administration are positioned as adjacent or learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark ETL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, forecasting, and validation workflows.
- Designed AWS and big-data-oriented data platform components using S3, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable transformation, querying, and stakeholder-facing reporting.
- Developed large-scale batch processing workflows using Python, SQL, PySpark, shell-style automation patterns, partition-aware design, Parquet-oriented storage, and historical retention models.
- Supported Hadoop/Hive-style data engineering patterns through distributed processing concepts, schema discipline, SQL transformations, curated layers, and analytical reporting structures.
- Built data quality, reconciliation, and validation practices across BMC TrueSight, CA Wily, AppDynamics, Oracle, and cloud-backed reporting layers to improve completeness, consistency, and trust in pipeline outputs.
- Performed performance analysis and troubleshooting by tracing source feeds, transformation logic, SQL behavior, failed outputs, latency issues, schema drift indicators, and downstream reporting discrepancies.
- Optimized SQL queries, Oracle schemas, Redshift-backed workloads, indexes, views, partitioning patterns, and reporting data models for performance, maintainability, and cost-aware processing.
- Collaborated with infrastructure, application, analytics, and business stakeholders to define data requirements, deliver reliable pipelines, document runbooks, and support enterprise reporting needs.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Developed analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.

- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and communicate findings to engineering teams.
- Built dashboards and operational reporting views that converted raw monitoring data into actionable performance and reliability insights.
- Supported production issue analysis by correlating monitoring signals, transaction behavior, and system performance data.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Managed enterprise telemetry environments with 50+ Enterprise Managers and 4,000–6,000 instrumented agents, supporting monitoring, operational analytics, dashboards, and production reliability.
- Built Perl and Korn shell automation for extracting, transforming, and distributing performance data, replacing manual exports with repeatable reporting workflows.
- Designed dashboards, alert logic, SLA thresholds, technical documentation, and reusable reporting standards adopted across client environments.
- Partnered with J2EE, WebLogic, infrastructure, database, and operations teams to analyze data, troubleshoot bottlenecks, and resolve production issues.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed enterprise telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection constraints.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, test findings, regression thresholds, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Designed and optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, Korn Shell, C++, Java

Flagship Projects

Serverless Lakehouse Platform (AWS)

Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, query optimization, reporting access, and validation checkpoints.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

HorizonScale — AI Capacity Forecasting Engine

Built a forecasting-oriented analytics workflow using Python, PySpark, Prophet, scikit-learn, and Streamlit to process infrastructure telemetry, prepare model-ready datasets, validate pipeline outputs, and deliver capacity insights.

Technologies: Python, PySpark, SQL, Prophet, scikit-learn, Streamlit, Data Validation

AI-Powered Job Search Pipeline

Built an AI-assisted Python automation pipeline with semantic duplicate detection, LLM-based scoring, JSON artifact generation, quality gates, document rendering, and application tracking using Claude Code, Grok/xAI APIs, FAISS, and sentence-transformers.

Technologies: Python, Claude Code, Grok / xAI API, FAISS, Sentence Transformers, JSON, Quality Gates